Thermal radiation and color

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use peak wavelength as one piece of evidence about a thermal spectrum.

SUCCESS CRITERIA - I can...

1. Relate higher temperature to shorter peak wavelength.
2. Use kelvin in Wien's law.
3. Distinguish spectral peak from perceived color.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "blackbody" and "peak" to explain your model or result. Sentence starters are optional.

blackbody: an ideal absorber/emitter used in a thermal-radiation model

peak: the location of the largest value on a specified graph

kelvin: the absolute temperature scale used in thermal ratios

01 Bring back an earlier idea

A light ray strikes a plane mirror at 20° from the normal. **Find** the reflection angle and label both angles on a sketch.

Word help: reflection = a wave returning from a boundary

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\lambda_{max}\,T = 2.9\times10^{-3}$ m·K.
Ideal blackbody; T in kelvin, λ_{max} in meters. A shorter peak wavelength means higher temperature, not one exact visible color.

Worked reasoning: A thermal spectrum's peak shifts toward shorter wavelength as absolute temperature rises. Wien's relation $\lambda_{peak}\,T \approx 2.9\times10^{-3}$ m·K gives $\lambda_{peak} \approx 9.7$ micrometers at 300 K.

Your turn: Sketch two broad thermal spectra, labeling axes and the hotter peak's shift.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Look and explain

Two ideal thermal emitters are at 3000 and 6000 K. Which has the shorter peak wavelength? Does that mean they emit only one color?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

04 Build the solution

Supplied peaks: 3000 K gives about 970 nm; 6000 K gives about 480 nm.
Compare wavelengths and identify which peak is outside visible light (roughly 400-700 nm).

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

An object has an infrared peak but looks red-hot.
Explain how both can be true.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

An upward rope pulse is 4 cm high before reaching an ideal free end. Is the reflected displacement upright or inverted?
Sketch it.

Word help: ideal = using stated simplifying conditions

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: 1 2 3

07 Exit check

Absolute temperature doubles. How does peak wavelength change?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

If temperature doubles, compare peak wavelengths.
Explain why a room-temperature object can radiate even when it is not visibly glowing.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3